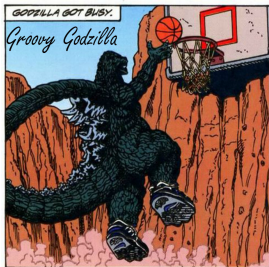


Predicting March Madness Basketball Games

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Introduction

- March Madness is a 68-team, single-elimination tournament across both the Men's and Women's divisions to determine the NCAA National Champions.
- The Kaggle competition (March Machine Learning Mania) evaluates the accuracy of predictions using the Brier score (BS).

$$BS = \frac{1}{N} \sum_{t=1}^N (f_t - o_t)^2 \quad (1)$$

- Where f_t is our prediction and o_t is the outcome.
- o_t is 1 if the team wins, and 0 if they lose.

Introduction

- A naive 'coin flip' strategy (50% probability for all games) yields a Brier Score of 0.25, serving as our baseline.

Example

If we predict that Duke has a win probability of 0.70 against Houston.

If Duke **wins**,

$$\text{Cost} = (1 - 0.70)^2 = (0.3)(0.3) = 0.09$$

If Duke **loses**,

$$\text{Cost} = (0 - 0.70)^2 = (0.7)(0.7) = 0.49$$

- We will call the two teams we are predicting Team A and team B.

- We looked at three different modeling techniques:
 - Two models gave us the probability outright (Logistic/XGBoost).
 - One model predicted the final scores (Simulation).
- We will be looking at the model that predicts final scores. We separate the men's and women's divisions.
- We are predicting the points that Team A will score against team B and vice versa. Then simulate +50,000 times.
- The ratio that Team A wins of the simulation is the predicted probability.

Men's Model

- Offensive Rating ($OffRating_A$)
- Defensive Rating ($DefRating_B$)
- Team Tempo ($AdjTempo_A$)
- Opponent Tempo ($AdjTempo_B$)
- Net Rating Difference
($NetRating_A - NetRating_B$)
- Elo Difference ($Elo_A - Elo_B$)

Women's Model

- Offensive Rating ($OffRating_A$)
- Defensive Rating ($DefRating_B$)
- Team Tempo ($AdjTempo_A$)
- Opponent Tempo ($AdjTempo_B$)
- Elo Difference ($Elo_A - Elo_B$)

Simulation Example

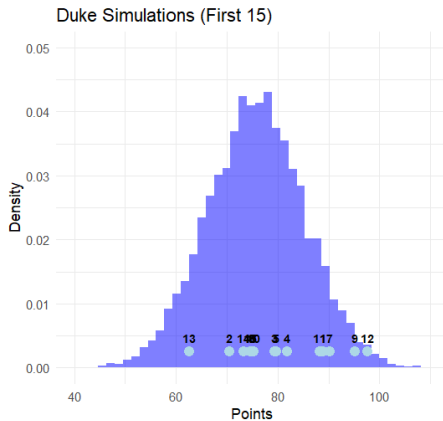


Figure 1: Duke Distribution Against Houston, $\sigma = 9.523$

Simulation Example

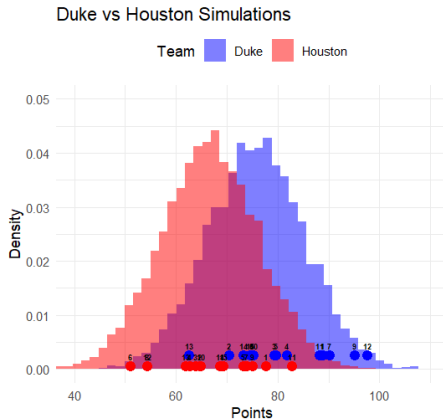


Figure 2: Duke vs Houston, $\sigma = 9.523$

- Model Achieved a Brier Score of **0.1305** (500 spots away from bronze).
- Men's tournament Brier score was **0.1448**. Women's was **0.1157**.
- Winning scores for the past three years were:

2023	2024	2025
0.1737	0.0531	0.1041

Table 1: Winning Brier Scores

2025 Results

- The 2025 Model was 79.9% accurate in picking the winners and losers.

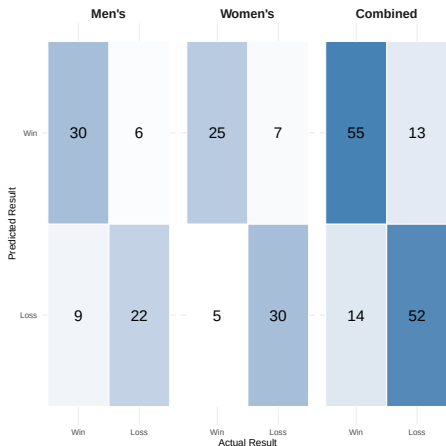


Figure 3: Confusion Matrix for 2025

2026 Results

- Prior to the tournament, we made 32 overrides to the models predictions. These were to give 100% probability to obvious favorites.
- Got 30 overrides correct, missed 2.
- Correctly picked that Michigan and UCLA would win.

Overrides	Original	Winner
0.13237	0.12797	0.10975

Table 2: 2026 Brier Score Comparisons

- These probabilities were 63.6%, 64.2%, 66.9% correct.

2026 Results

- The 2026 Model was 77.8% accurate.

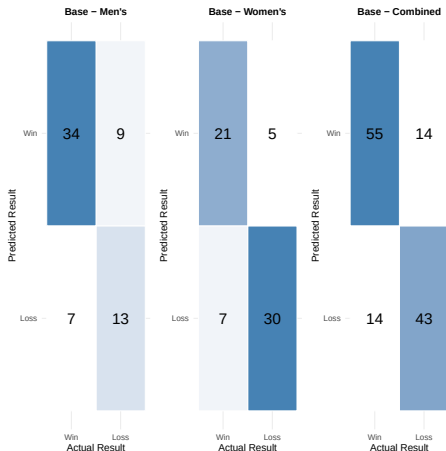


Figure 4: Confusion Matrix for 2026

2026 Worst Predictions

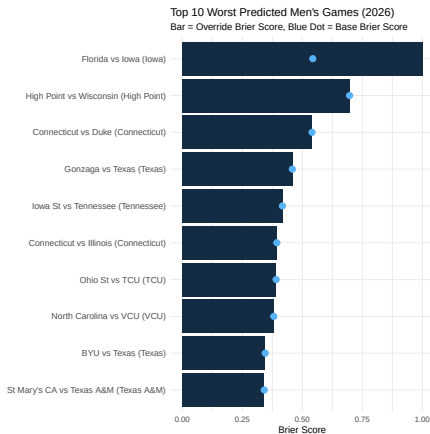


Figure 5: Men's BS Graph

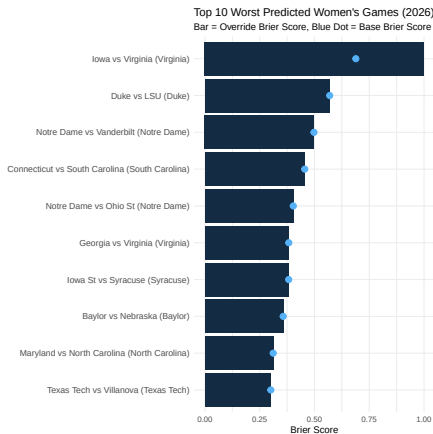


Figure 6: Women's BS Graph

Thank You!